$$\begin{bmatrix} 1 & -3 & 2 & -4 & 0 \\ 0 & 0 & 5 & -7 & 0 \\ 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 5 & 0 \end{bmatrix} \xrightarrow[-\text{Row}3]{\text{Row}4=\text{Row}4} \begin{bmatrix} 1 & -3 & 2 & -4 & 0 \\ 0 & 0 & 5 & -7 & 0 \\ 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\times 1/5]{\text{Row}3=\text{Row}3}$$

$$\begin{bmatrix} \boxed{1} & -3 & 2 & -4 & 0 \\ 0 & 0 & \boxed{5} & -7 & 0 \\ 0 & 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

. It's easy to verify that the final matrix is in echelon form. Row 1 Column 1, Row 2 Column 3 and Row 3 Column 4 are pivots. ∴ Cols 1, 3 and 4 are pivot cols.

It's easy to see that the subspace spanned by the given vectors is Col A. By Theorem 13, the basis of Col A is 1, 3, 4th columns, i.e.

$$\left\{ \begin{bmatrix} 1 \\ -3 \\ 2 \\ -4 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} -4 \\ 5 \\ -5 \\ 7 \end{bmatrix} \right\} \square$$

14. $\begin{bmatrix} 1 \\ -1 \\ -2 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ -1 \\ 6 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -6 \\ 8 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \\ -7 \\ 7 \end{bmatrix}, \begin{bmatrix} 3 \\ -8 \\ 9 \\ -5 \end{bmatrix}$

Solution: We shall repeat the procedure of Problem 13.

$$\begin{bmatrix} 1 & 2 & 0 & -1 & 3 & 0 \\ -1 & -3 & 2 & 4 & -8 & 0 \\ -2 & -1 & -6 & -7 & 9 & 0 \\ 5 & 6 & 8 & 7 & -5 & 0 \end{bmatrix} \xrightarrow[+\text{Row}1]{\text{Row}2=\text{Row}2} \begin{bmatrix} 1 & 2 & 0 & -1 & 3 & 0 \\ 0 & -1 & 2 & 3 & -5 & 0 \\ -2 & -1 & -6 & -7 & 9 & 0 \\ 5 & 6 & 8 & 7 & -5 & 0 \end{bmatrix}$$

$$\xrightarrow[2\times\text{Row}1]{\text{Row}3=\text{Row}3+} \begin{bmatrix} 1 & 2 & 0 & -1 & 3 & 0 \\ 0 & -1 & 2 & 3 & -5 & 0 \\ 0 & 3 & -6 & -9 & 15 & 0 \\ 5 & 6 & 8 & 7 & -5 & 0 \end{bmatrix} \xrightarrow[-5\times\text{Row}1]{\text{Row}4=\text{Row}4} \begin{bmatrix} 1 & 2 & 0 & -1 & 3 & 0 \\ 0 & -1 & 2 & 3 & -5 & 0 \\ 0 & 3 & -6 & -9 & 15 & 0 \\ 0 & -4 & 8 & 12 & -20 & 0 \end{bmatrix}$$

$$\xrightarrow[+3\times\text{Row}2]{\text{Row}3=\text{Row}3} \begin{bmatrix} 1 & 2 & 0 & -1 & 3 & 0 \\ 0 & -1 & 2 & 3 & -5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -4 & 8 & 12 & -20 & 0 \end{bmatrix} \xrightarrow[-4\times\text{Row}2]{\text{Row}4=\text{Row}4} \begin{bmatrix} \boxed{1} & 2 & 0 & -1 & 3 & 0 \\ 0 & \boxed{-1} & 2 & 3 & -5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Row 1 Col 1 and Row 2 Col 2 are pivots. ∴ The basis of Col A is

$$\left\{ \begin{bmatrix} 1 \\ -1 \\ -2 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ -1 \\ 6 \end{bmatrix} \right\} \square$$